ARIMA process

Ithuteng Ntakhwana ID: 201801847

2024-04-08

We will be exploring the Airline passengers dataset in R and fitting 3 ARIMA models

```
#load the libraries
library(forecast)
```

```
## Warning: package 'forecast' was built under R version 4.2.3
```

```
## Registered S3 method overwritten by 'quantmod':
##   method            from
##   as.zoo.data.frame zoo
```

```
library(ggplot2)
```

```
## Warning: package 'ggplot2' was built under R version 4.2.3
```

```
library(dplyr)
```

```
## Warning: package 'dplyr' was built under R version 4.2.3
```

```
##
## Attaching package: 'dplyr'
```

```
## The following objects are masked from 'package:stats':
##
##     filter, lag
```

```
## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
```

```
#Load Air passengers dataset embedded in R
passengers = data.frame(year = rep(1949:1960, each = 12),
                        month = c(1:12,1:12,1:12,1:12,1:12,1:12,1:12,1:12,1:12,1:12,1:12,1:12),
                        fq = c(112, 118, 132, 129, 121, 135, 148, 148, 136, 119, 104, 118,
                                 115, 126, 141, 135, 125, 149, 170, 170, 158, 133, 114, 140,
                                 145, 150, 178, 163, 172, 178, 199, 199, 184, 162, 146, 166,
                                 171, 180, 193, 181, 183, 218, 230, 242, 209, 191, 172, 194,
                                 196, 196, 236, 235, 229, 243, 264, 272, 237, 211, 180, 201,
```

```
                                204, 188, 235, 227, 234, 264, 302, 293, 259, 229, 203, 229,
                                242, 233, 267, 269, 270, 315, 364, 347, 312, 274, 237, 278,
                                284, 277, 317, 313, 318, 374, 413, 405, 355, 306, 271, 306,
                                315, 301, 356, 348, 355, 422, 465, 467, 404, 347, 305, 336,
                                340, 318, 362, 348, 363, 435, 491, 505, 404, 359, 310, 337,
                                360, 342, 406, 396, 420, 472, 548, 559, 463, 407, 362, 405,
                                417, 391, 419, 461, 472, 535, 622, 606, 508, 461, 390, 432
                        ))

#first rows of passengers dataset
head(passengers)
```

```
##   year month  fq
## 1 1949     1 112
## 2 1949     2 118
## 3 1949     3 132
## 4 1949     4 129
## 5 1949     5 121
## 6 1949     6 135
```

```
#stucture of dataset
str(passengers)
```

```
## 'data.frame':    144 obs. of  3 variables:
##  $ year : int  1949 1949 1949 1949 1949 1949 1949 1949 1949 1949 ...
##  $ month: int  1 2 3 4 5 6 7 8 9 10 ...
##  $ fq   : num  112 118 132 129 121 135 148 148 136 119 ...
```

```
#summary of dataset
summary(passengers)
```

```
##       year          month             fq
##  Min.   :1949   Min.   : 1.00   Min.   :104.0
##  1st Qu.:1952   1st Qu.: 3.75   1st Qu.:180.0
##  Median :1954   Median : 6.50   Median :265.5
##  Mean   :1954   Mean   : 6.50   Mean   :280.3
##  3rd Qu.:1957   3rd Qu.: 9.25   3rd Qu.:360.5
##  Max.   :1960   Max.   :12.00   Max.   :622.0
```

```
#check for missing values
library(zoo)
```

```
## Warning: package 'zoo' was built under R version 4.2.3
```

```
##
## Attaching package: 'zoo'
```

```
## The following objects are masked from 'package:base':
##
##     as.Date, as.Date.numeric
```

```
passengers = na.approx(passengers)



#summary of the air plane passengers dataset, mean, median etc
summary(passengers)
```

```
##       year          month             fq
##  Min.   :1949   Min.   : 1.00   Min.   :104.0
##  1st Qu.:1952   1st Qu.: 3.75   1st Qu.:180.0
##  Median :1954   Median : 6.50   Median :265.5
##  Mean   :1954   Mean   : 6.50   Mean   :280.3
##  3rd Qu.:1957   3rd Qu.: 9.25   3rd Qu.:360.5
##  Max.   :1960   Max.   :12.00   Max.   :622.0
```

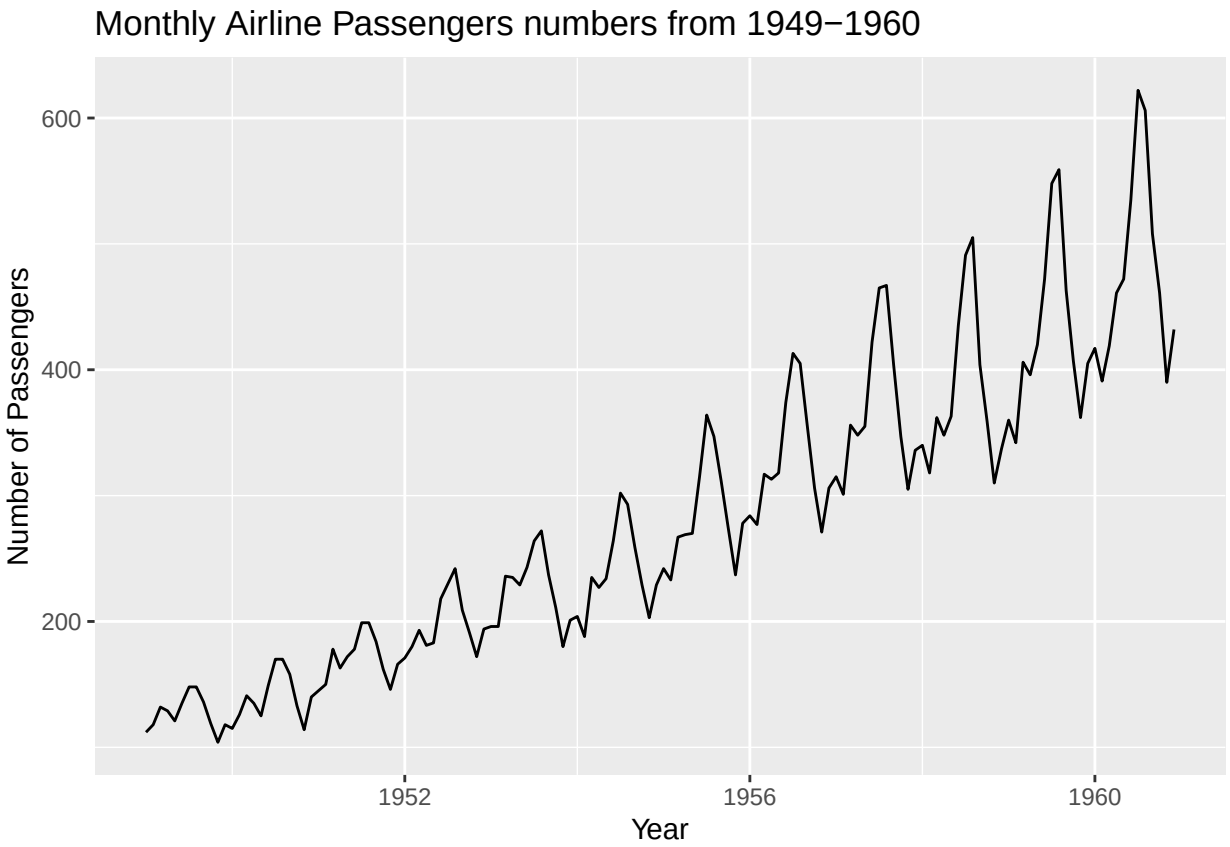
Explanatory Data Analysis of the Time Series

```
#Creating Time Series object
passengers_time = ts(passengers[,-c(1,2)], frequency = 12,start = c(1949,1))
passengers_time
```

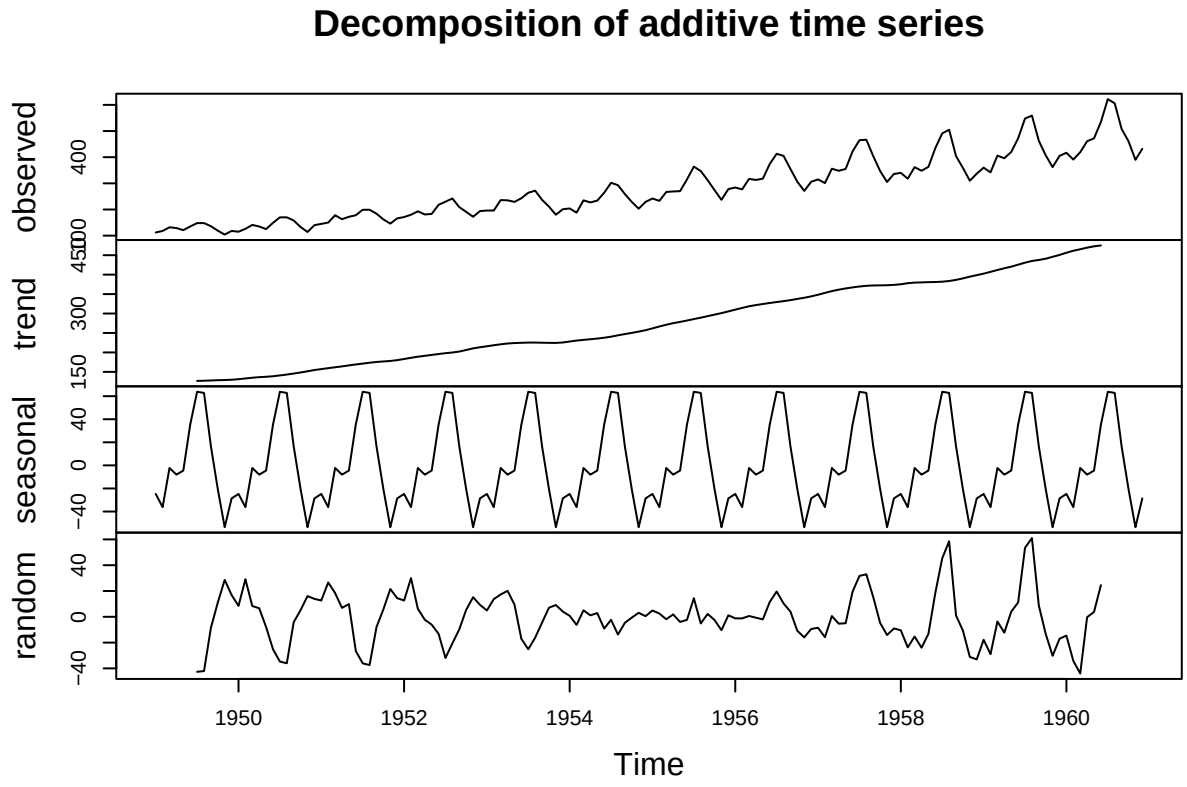
```
##      Jan Feb Mar Apr May Jun Jul Aug Sep Oct Nov Dec
## 1949 112 118 132 129 121 135 148 148 136 119 104 118
## 1950 115 126 141 135 125 149 170 170 158 133 114 140
## 1951 145 150 178 163 172 178 199 199 184 162 146 166
## 1952 171 180 193 181 183 218 230 242 209 191 172 194
## 1953 196 196 236 235 229 243 264 272 237 211 180 201
## 1954 204 188 235 227 234 264 302 293 259 229 203 229
## 1955 242 233 267 269 270 315 364 347 312 274 237 278
## 1956 284 277 317 313 318 374 413 405 355 306 271 306
## 1957 315 301 356 348 355 422 465 467 404 347 305 336
## 1958 340 318 362 348 363 435 491 505 404 359 310 337
## 1959 360 342 406 396 420 472 548 559 463 407 362 405
## 1960 417 391 419 461 472 535 622 606 508 461 390 432
```

```
#plot of Time Series
ggplot(data = data.frame(Date = time(passengers_time),
                         Passengers = passengers_time),
                         aes(x = Date, y = Passengers)) + geom_line() +
                        labs(title = "Monthly Airline Passengers numbers from 1949-1960",
                             x = "Year",
                             y = "Number of Passengers")
```

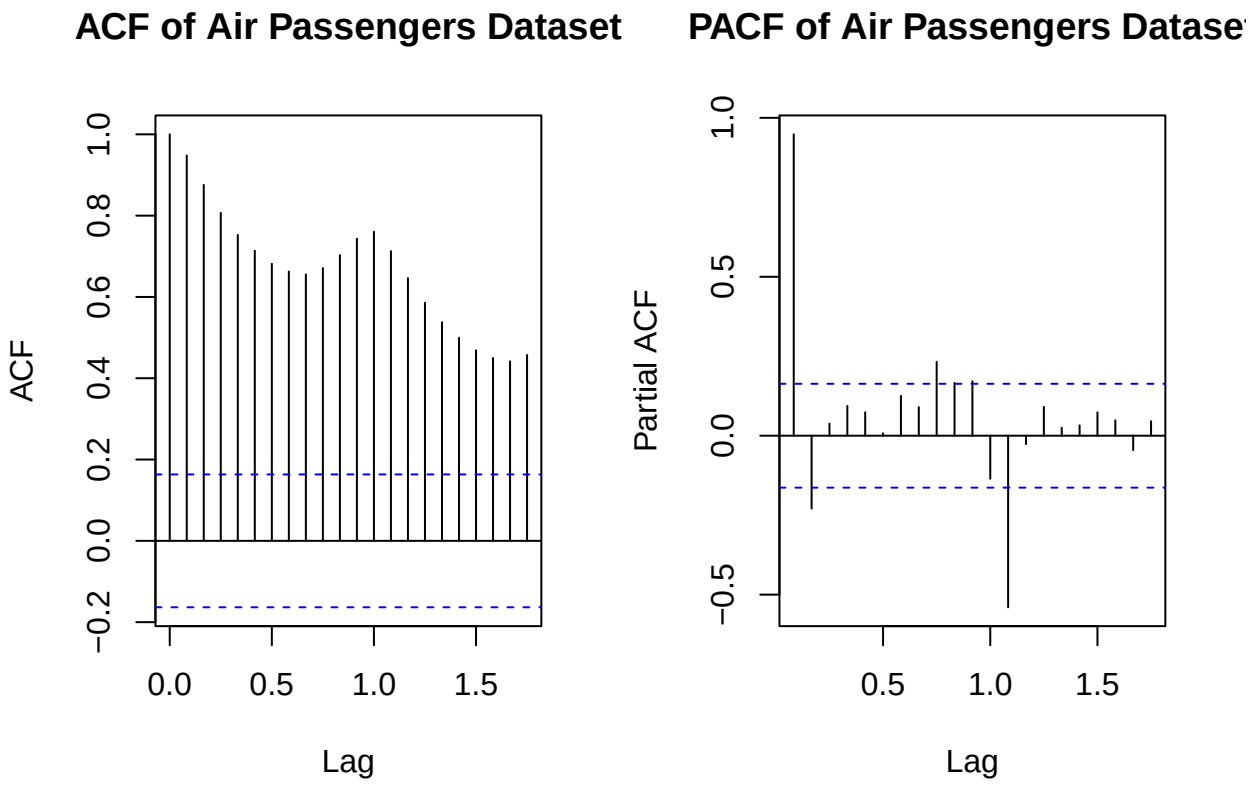
```
## Don't know how to automatically pick scale for object of type <ts>. Defaulting
## to continuous.
## Don't know how to automatically pick scale for object of type <ts>. Defaulting
## to continuous.
```




```
#Finding the trend and seasonal components
dec = decompose(passengers_time)
plot(dec)
```

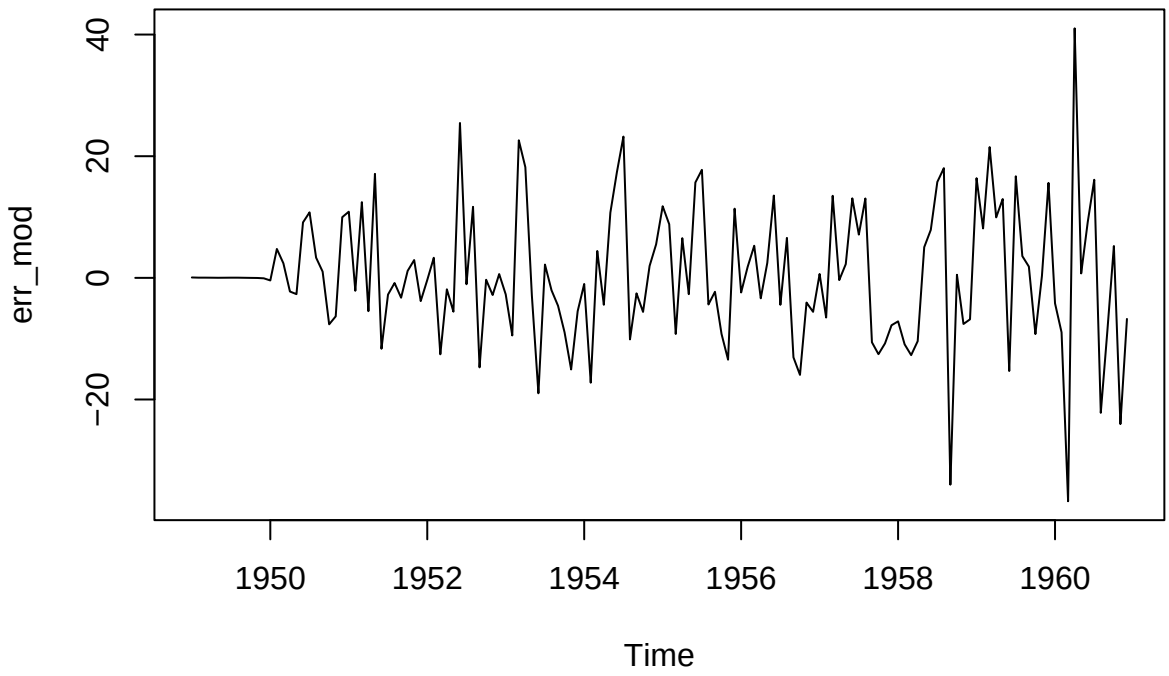



```
#ARIMA(p,q) components
#Autocorrelation function (ACF)
#Partial Autocorrelation function (PACF)
par(mfrow = c(1,2))
acf(passengers_time, main = "ACF of Air Passengers Dataset")
pacf(passengers_time,main = "PACF of Air Passengers Dataset")
```

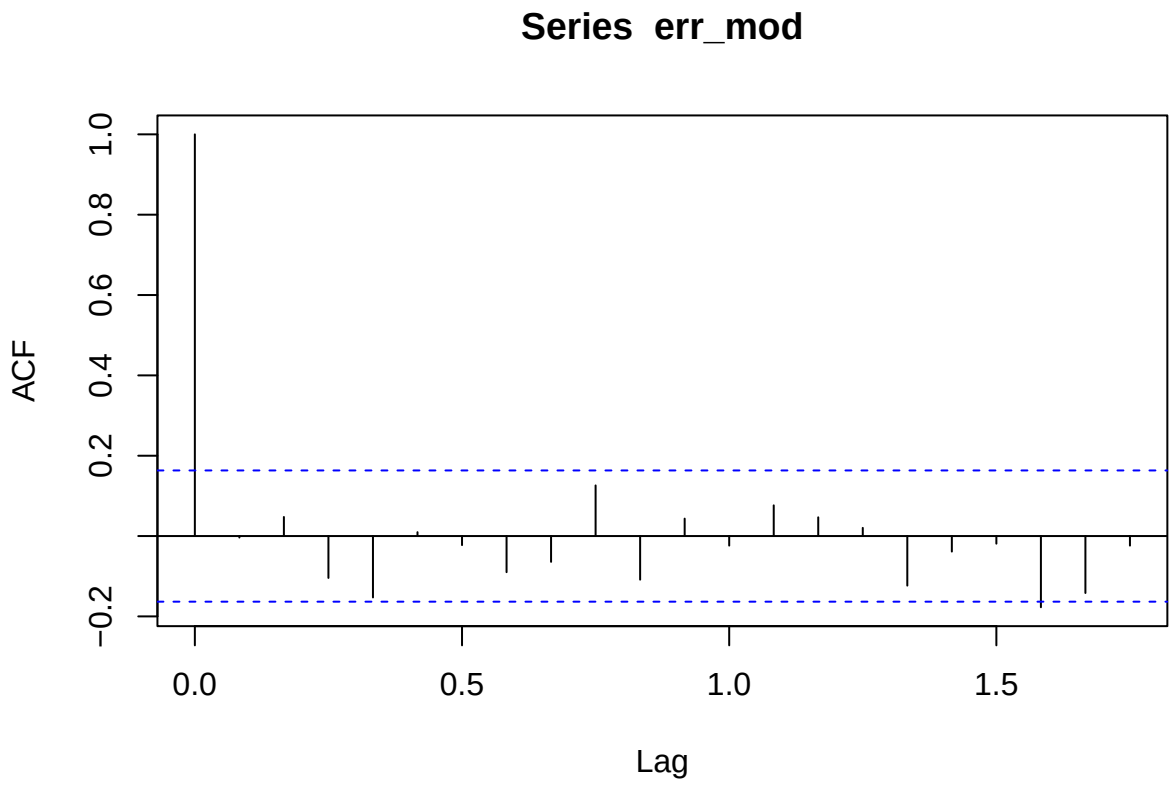



```
#Fit ARIMA model ARIMA(0,1,1)
ar_model = arima(passengers_time, order = c(0,1,1),seasonal = list(order = c(0,1,1) ,period = 12))

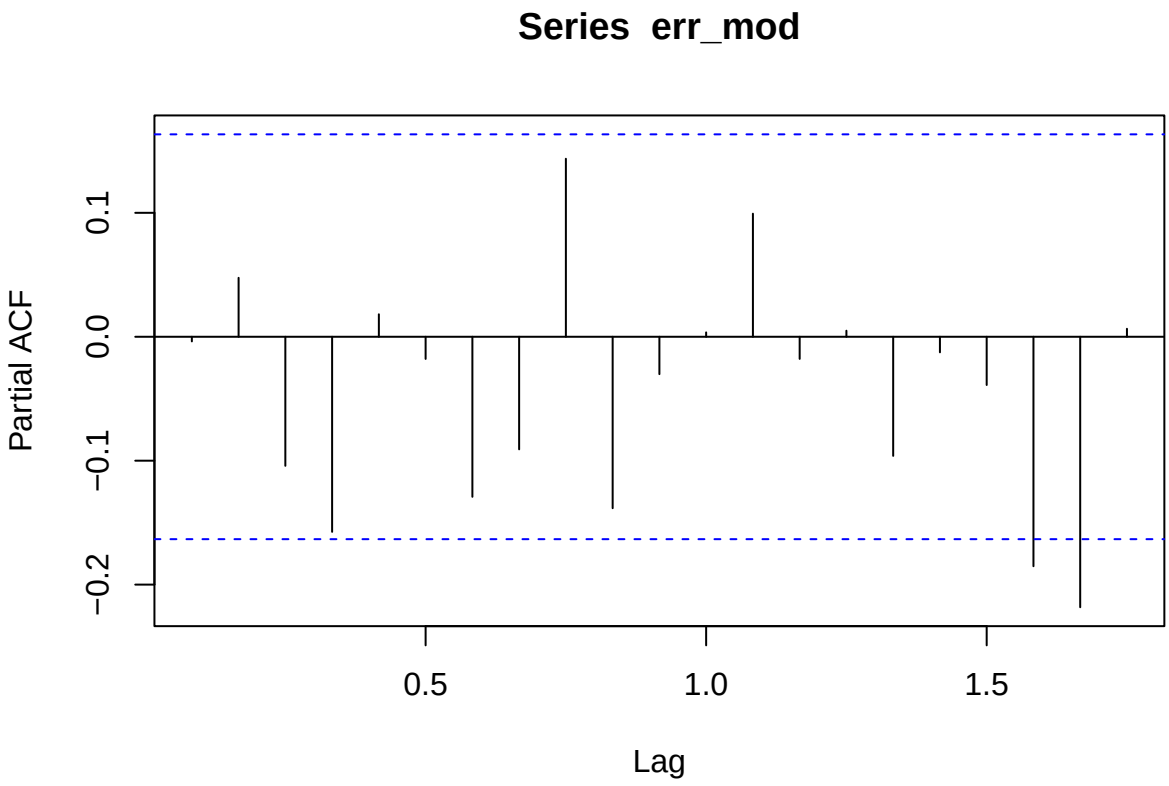
#Plotting residuals for ARIMA (0,1,1)
err_mod = resid(ar_model)
plot(err_mod)
```




```
acf(err_mod)
```




```
pacf(err_mod)
```



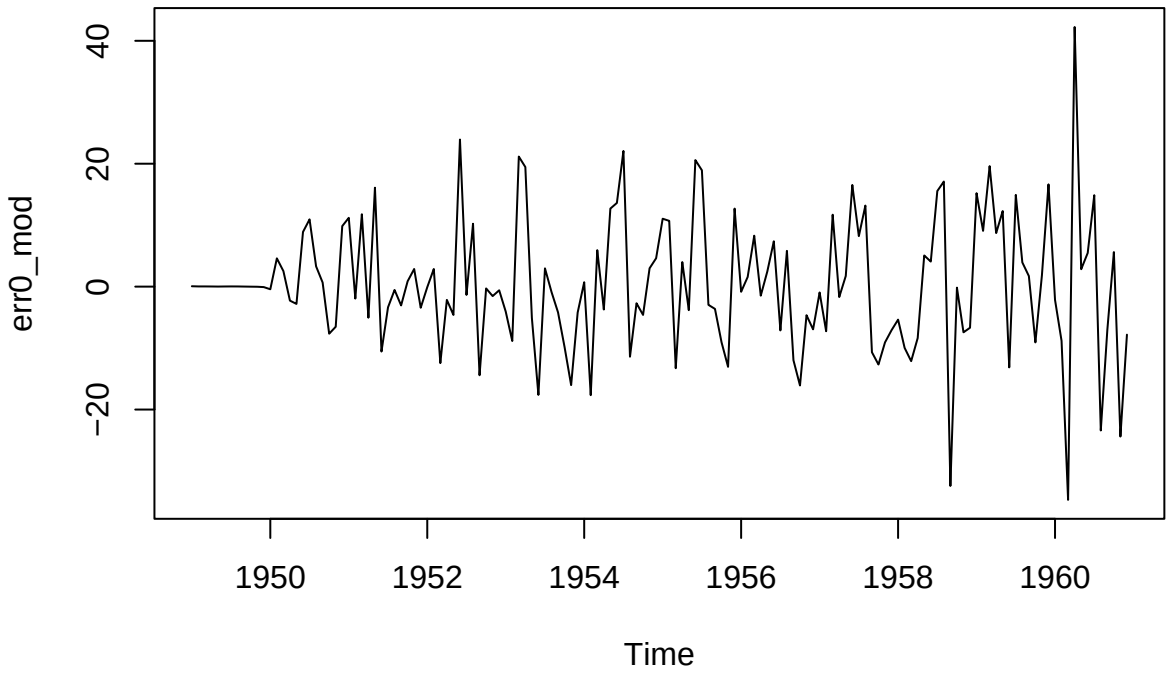

```
#summary of ARIMA(0,1,1) model
summary(ar_model)
```

```
##
## Call:
## arima(x = passengers_time, order = c(0, 1, 1), seasonal = list(order = c(0,
##     1, 1), period = 12))
##
## Coefficients:
##           ma1     sma1
##       -0.3087  -0.1074
## s.e.   0.0890   0.0828
##
## sigma^2 estimated as 135.4:  log likelihood = -507.5,  aic = 1021
##
## Training set error measures:
##                      ME     RMSE      MAE        MPE     MAPE      MASE
## Training set 0.3235529 11.09952 8.162425 0.04409006 2.897135 0.3156373
##                      ACF1
## Training set -0.003747514
```

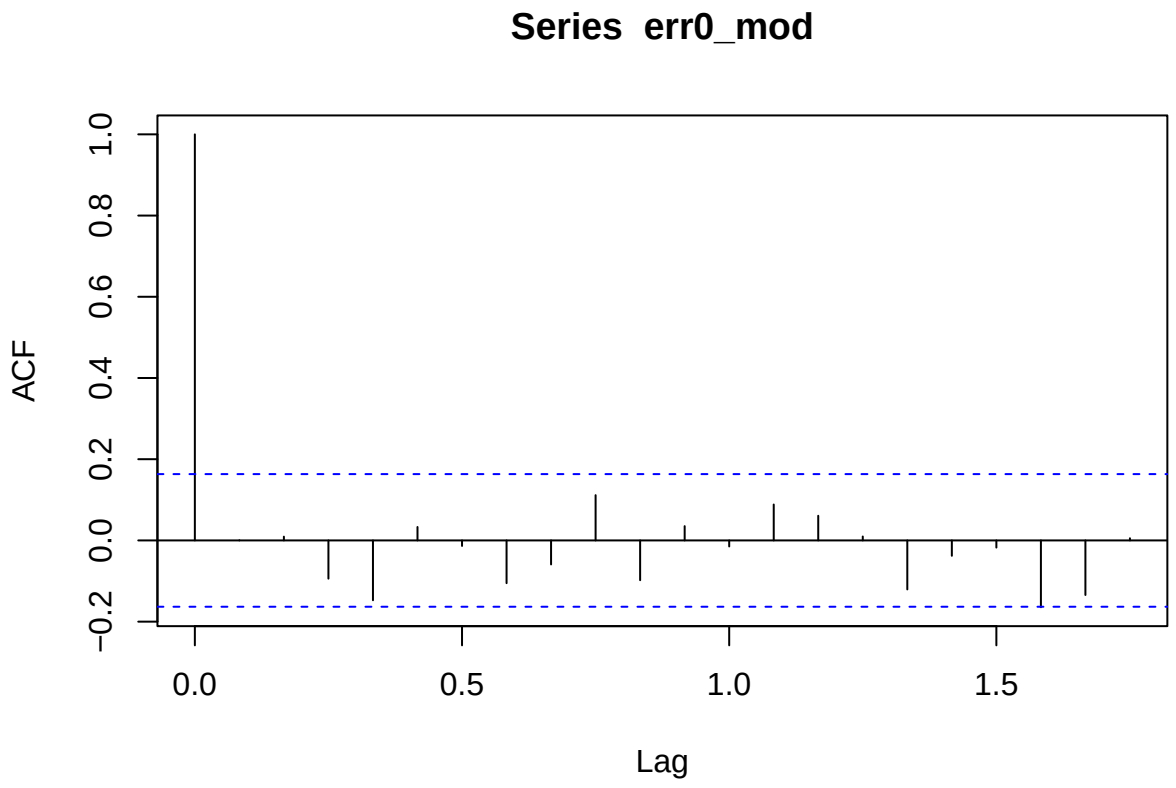
```
#Fit ARIMA model ARIMA(1,1,1)
ar0_model = arima(passengers_time, order = c(1,1,1),seasonal = list(order = c(1,1,1) ,period = 12))

#Plotting residuals for ARIMA (1,1,1)
```

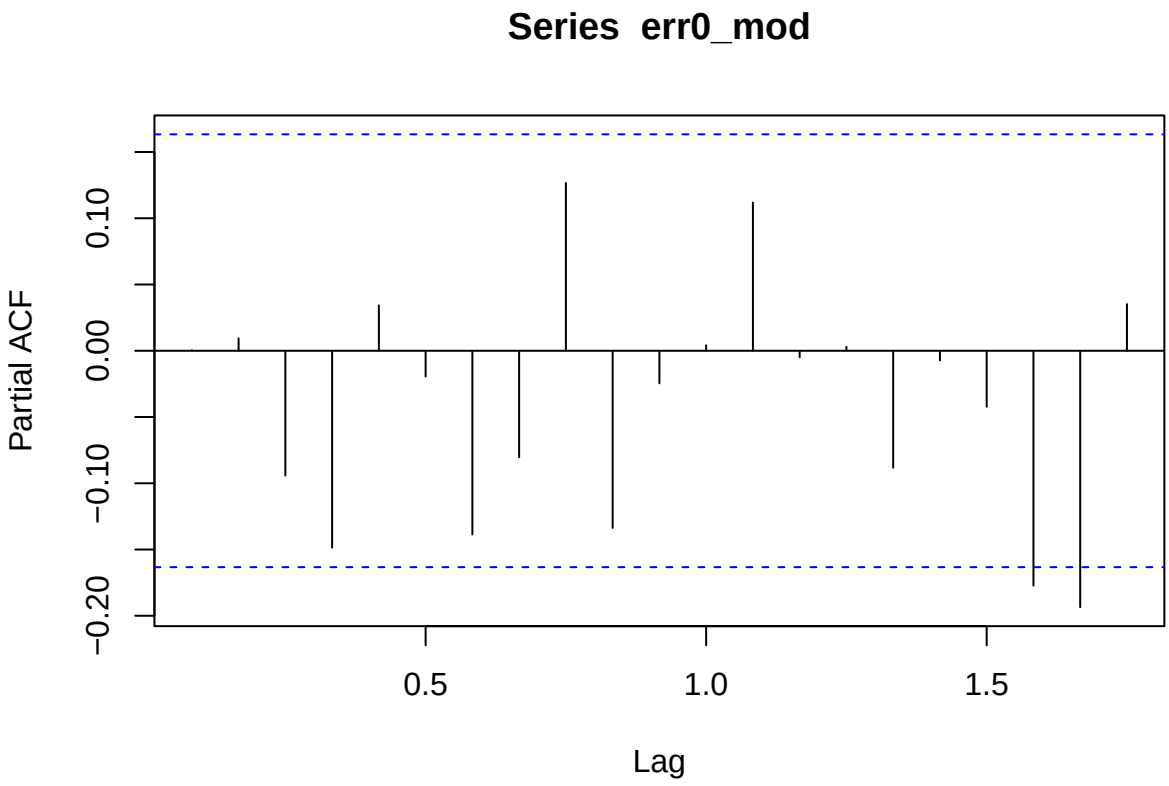
```
err0_mod = resid(ar0_model)
plot(err0_mod)
```




```
acf(err0_mod)
```




```
pacf(err0_mod)
```




```
#summary of ARIMA(1,1,1) model
summary(ar0_model)
```

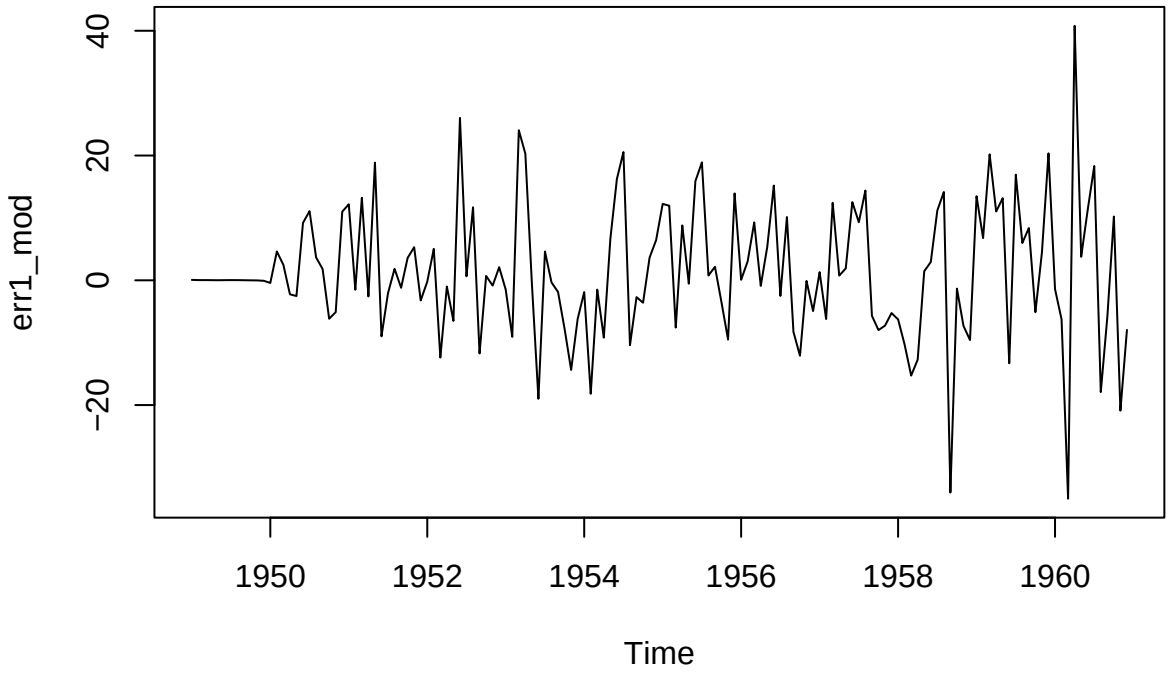
```
##
## Call:
## arima(x = passengers_time, order = c(1, 1, 1), seasonal = list(order = c(1,
##     1, 1), period = 12))
##
## Coefficients:
##           ar1      ma1     sar1    sma1
##       -0.1387  -0.2027  -0.9228  0.8329
## s.e.   0.5860   0.6123   0.2386  0.3519
##
## sigma^2 estimated as 130.8:  log likelihood = -506.15,  aic = 1022.3
##
## Training set error measures:
##                      ME     RMSE      MAE        MPE     MAPE      MASE
## Training set 0.3152762 10.91023 8.003276 0.04379133 2.848878 0.3094831
##                      ACF1
## Training set 0.0004688507
```

```
#fitting an auto ARIMA model
ar1_model = auto.arima(passengers_time)
ar1_model
```

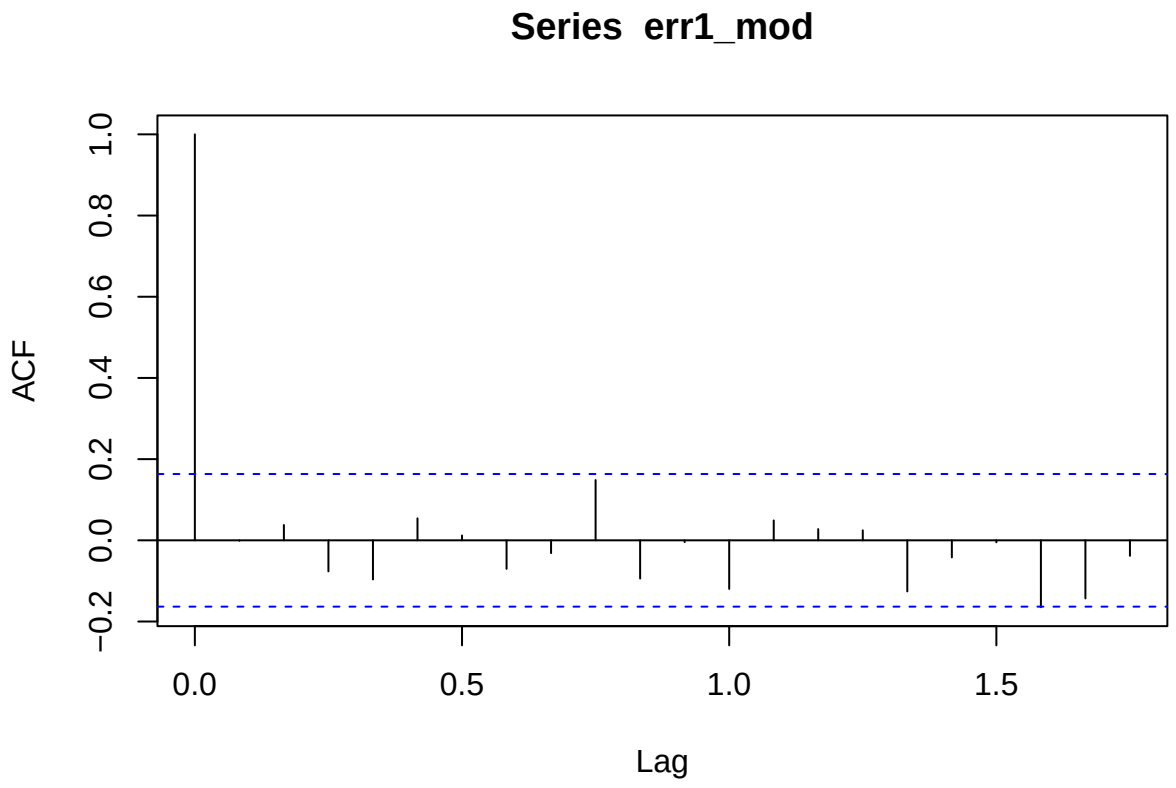
```
## Series: passengers_time
```

```
## ARIMA(2,1,1)(0,1,0)[12]
##
## Coefficients:
##          ar1     ar2      ma1
##       0.5960  0.2143  -0.9819
## s.e.  0.0888  0.0880   0.0292
##
## sigma^2 = 132.3:  log likelihood = -504.92
## AIC=1017.85   AICc=1018.17   BIC=1029.35
```

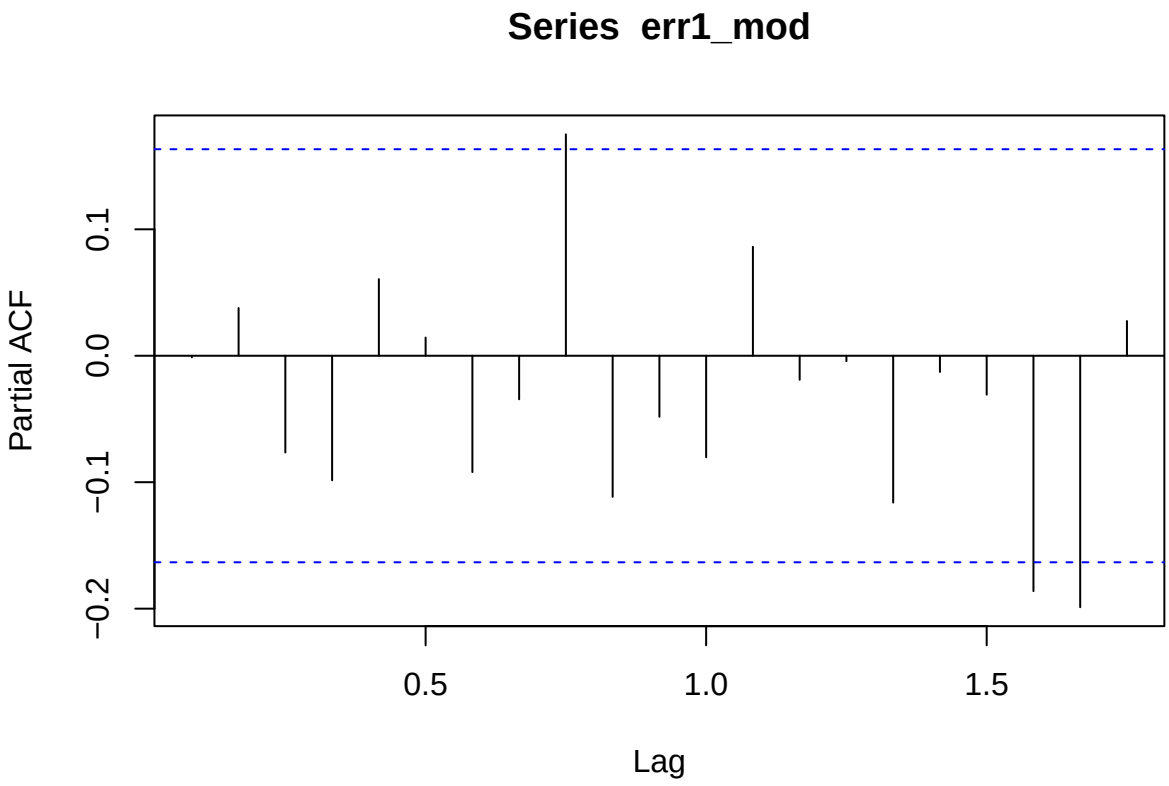
```
#Plotting residuals for auto ARIMA model
err1_mod = resid(ar1_model)
plot(err1_mod)
```




```
acf(err1_mod)
```




```
pacf(err1_mod)
```




```
#summary of auto ARIMA model
summary(ar1_model)
```

```
## Series: passengers_time
## ARIMA(2,1,1)(0,1,0)[12]
##
## Coefficients:
##          ar1     ar2      ma1
##       0.5960  0.2143  -0.9819
## s.e.  0.0888  0.0880   0.0292
##
## sigma^2 = 132.3:  log likelihood = -504.92
## AIC=1017.85   AICc=1018.17   BIC=1029.35
##
## Training set error measures:
##                     ME     RMSE     MAE      MPE     MAPE     MASE        ACF1
## Training set 1.3423 10.84619 7.86754 0.420698 2.800458 0.245628 -0.00124847
```

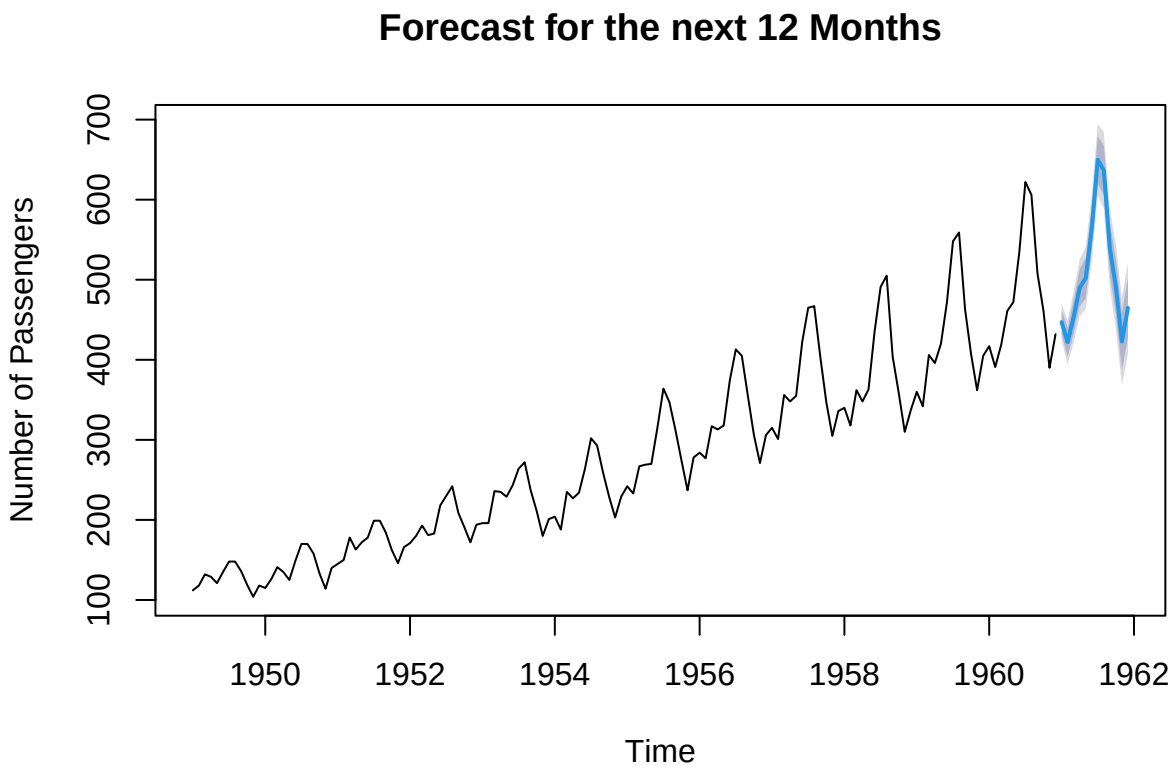
```
#forecasting for ARIMA(0,1,1) model

forcast_v0 = forecast(object = ar_model, h =12 )
forcast_v0
```

```
##          Point Forecast    Lo 80    Hi 80    Lo 95    Hi 95
## Jan 1961       447.0532 432.1396 461.9668 424.2448 469.8616
```

```
## Feb 1961       421.8774 403.7469 440.0080 394.1492 449.6057
## Mar 1961       453.5262 432.6692 474.3832 421.6281 485.4243
## Apr 1961       489.9008 466.6346 513.1670 454.3182 525.4834
## May 1961       502.1835 476.7352 527.6318 463.2637 541.1033
## Jun 1961       564.2246 536.7671 591.6822 522.2319 606.2173
## Jul 1961       649.7953 620.4658 679.1247 604.9398 694.6508
## Aug 1961       636.7146 605.6258 667.8035 589.1684 684.2609
## Sep 1961       538.9209 506.1670 571.6748 488.8282 589.0136
## Oct 1961       491.0672 456.7290 525.4055 438.5514 543.5831
## Nov 1961       422.8243 386.9716 458.6770 367.9923 477.6562
## Dec 1961       464.7526 427.4468 502.0583 407.6984 521.8067
```

```
#Next 12 months
plot(forcast_v0,
     main = "Forecast for the next 12 Months",
     xlab = "Time",
     ylab = "Number of Passengers")
```



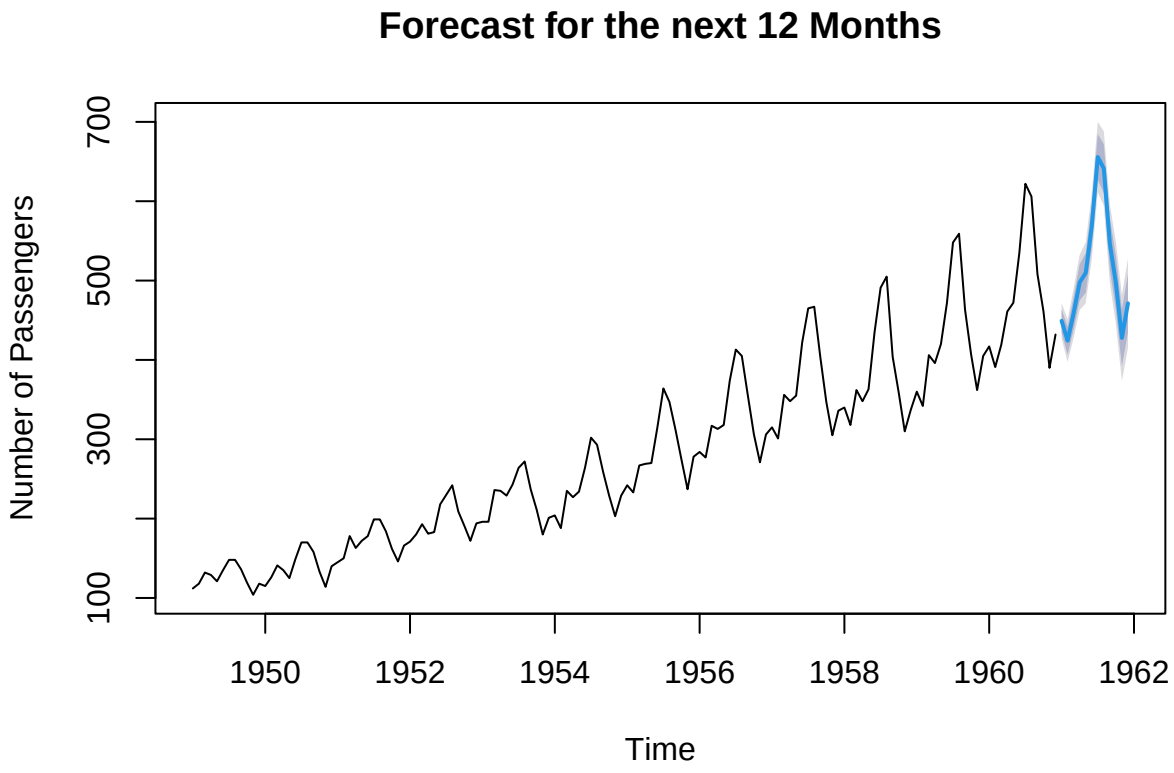

```
#forecasting for ARIMA(1,1,1) model

forcast_v1 = forecast(object = ar0_model, h =12 )
forcast_v1
```

```
##          Point Forecast    Lo 80    Hi 80    Lo 95    Hi 95
## Jan 1961       449.2223 434.5545 463.8901 426.7899 471.6547
```

```
## Feb 1961       424.3072 406.7452 441.8691 397.4485 451.1658
## Mar 1961       459.0023 438.6160 479.3886 427.8241 490.1805
## Apr 1961       497.7220 474.9011 520.5430 462.8204 532.6237
## May 1961       509.7462 484.7209 534.7715 471.4733 548.0190
## Jun 1961       568.1989 541.1491 595.2488 526.8297 609.5681
## Jul 1961       655.7364 626.8032 684.6696 611.4868 699.9859
## Aug 1961       641.1725 610.4712 671.8737 594.2190 688.1259
## Sep 1961       546.3741 514.0013 578.7470 496.8642 595.8841
## Oct 1961       496.7500 462.7878 530.7123 444.8092 548.6908
## Nov 1961       427.6964 392.2158 463.1769 373.4335 481.9592
## Dec 1961       471.2915 434.3550 508.2281 414.8019 527.7811
```

```
#Next 12 months
plot(forcast_v1,
     main = "Forecast for the next 12 Months",
     xlab = "Time",
     ylab = "Number of Passengers")
```




```
#forecasting for ARIMA(1,1,1) model

forcast_v2 = forecast(object = ar1_model, h =12 )
forcast_v2
```

```
##          Point Forecast    Lo 80    Hi 80    Lo 95    Hi 95
## Jan 1961       445.6349 430.8903 460.3795 423.0851 468.1847
```

```
## Feb 1961       420.3950 403.0907 437.6993 393.9304 446.8596
## Mar 1961       449.1983 429.7726 468.6240 419.4892 478.9074
## Apr 1961       491.8399 471.0270 512.6529 460.0092 523.6707
## May 1961       503.3945 481.5559 525.2330 469.9953 536.7937
## Jun 1961       566.8624 544.2637 589.4612 532.3007 601.4242
## Jul 1961       654.2602 631.0820 677.4383 618.8122 689.7081
## Aug 1961       638.5975 614.9704 662.2246 602.4630 674.7320
## Sep 1961       540.8837 516.9028 564.8647 504.2081 577.5594
## Oct 1961       494.1266 469.8624 518.3909 457.0177 531.2356
## Nov 1961       423.3327 398.8381 447.8273 385.8715 460.7939
## Dec 1961       465.5076 440.8229 490.1923 427.7556 503.2596
```

```r
#Next 12 months
plot(forcast_v2,
     main = "Forecast for the next 12 Months",
     xlab = "Time",
     ylab = "Number of Passengers")
```

